The Emergent Geometry Sub-Programme

Presentation Note 2

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Abstract

The emergent geometry sub-programme of the Cosmochrony corpus addresses a single central question: how does the admissible Weil–Heisenberg fibre give rise to an effective four-dimensional Lorentzian geometry? Starting from the admissibility filter Π_q acting on the Weil representation V_ρ of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, the sub-programme closes the following chain:

$$\Pi_q \implies V_\rho \simeq L^2(\mathbb{Z}/q\mathbb{Z}) \implies \mathrm{Heis}_3(\mathbb{R}) \implies L_{\mathrm{eff}} \implies g^{\mu\nu} \implies g^{\mu\nu} = 2\eta^{\mu\nu}.$$

This note maps eleven constituent papers (Q5a, Q5a-O2, Q5b, Q6b, Q7–Q11, U1, W1, H2) across five internal phases and records the status of every result. The remaining open items are not obstructions to the admissible-sector metric closure. They concern, respectively, the finite-q bridge formulation and the extension of the Q5a Mosco tightness control from the admissible sector to the full function space.

Keywords. emergent spacetime; Heisenberg group; Carnot geometry; Mosco convergence; Lorentzian signature; effective metric; sub-Riemannian geometry; Casimir rigidity; non-injective projection; admissibility; presentation note.

Contents

1 Motivation and Central Question

The emergent geometry sub-programme answers the following question.

Central question. The admissibility filter Π_q of the Cosmochrony framework acts on the Weil representation $V_\rho \simeq L^2(\mathbb{Z}/q\mathbb{Z})$ of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, selecting admissible modes according to the Born–Infeld bounded-flux constraint. *What effective geometry is selected in the large-q limit of this structure, and what determines the numerical values of its metric coefficients?*

The practical importance of this question is that the entire gravitational, gauge, and fermionic content of Branch III depends on the identification of an effective spacetime. Without a derived metric, the dynamical outputs of the Gravity paper [5], the gauge structure of Q6a [4], and the spinorial structure of Q14 [10] would each require an independently postulated background. The emergent geometry sub-programme removes this postulate: spacetime is a forced consequence of admissibility, not an input.

This note is concerned exclusively with the reconstruction of the effective metric $g^{\mu\nu} = 2\eta^{\mu\nu}$. The dynamical content — why the effective metric satisfies the Einstein equations — is the subject of the spectral gravity sub-programme and is not claimed here.

2 Position in the Cosmochrony Programme

The emergent geometry sub-programme occupies the interface between Branch I (axiomatic primitive) and Branch III (physical observables). It takes the algebraic output of Branch I — the Weil representation of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ and the Born–Infeld admissibility constraint — and produces the effective Lorentzian metric used throughout Branch III.

Its relation to the spectral admissibility sub-programme (Presentation Note 1) is the following. Note 1 establishes that the admissible sector is the spin-$\frac{1}{2}$ sector $V_\rho \cong \mathbb{C}^2$ and that $\Sigma_c(n_3) = 3$ with $\mathrm{Im}\,\mathbb{H} \cong \mathfrak{su}(2)$. The emergent geometry sub-programme takes these facts as inputs and asks: what metric structure do they support?

Remark 1 (Vocabulary note)**.** Throughout this note, the term *admissibility constraint* replaces the earlier "relaxation" vocabulary of the preliminary Cosmochrony formulation. The substrate χ is static: no substrate dynamics or relaxation mechanism is postulated at the foundational level. Observable evolution is an induced ordering of admissible projections, not a dynamics of χ itself.

Remark 2 (Scope: metric reconstruction only)**.** This sub-programme closes the chain $\Pi_q \to g^{\mu\nu} = 2\eta^{\mu\nu}$. It does not address the dynamics of $g^{\mu\nu}$: the question of why the effective metric satisfies the Einstein equations is handled by the spectral gravity sub-programme, which takes the metric established here as its starting point.

3 Logical Chain of the Sub-Programme

The sub-programme is organised around the following derivation chain.

$$\underbrace{\Pi_q}_{\text{admissibility filter}} \Longrightarrow \underbrace{L_\Pi = -A\partial_x^2}_{\substack{\text{1D shadow} \\ \text{(Mosco, Q5a)}}} \Longrightarrow \underbrace{L_{\text{eff}} \text{ on } \mathbb{R}_\tau \times \mathrm{Heis}_3(\mathbb{R})}_{\substack{\text{4D effective operator} \\ \text{(Carnot, Q5b)}}} \Longrightarrow \underbrace{g^{\mu\nu} \propto A_{\mu\nu}}_{\substack{\text{metric extraction} \\ \text{(Q5b + Q9)}}} \Longrightarrow \underbrace{g^{\mu\nu} = 2\eta^{\mu\nu}}_{\substack{\text{metric closure} \\ \text{(Q8–Q11, with U1, W1, H2)}}}. \tag{1}$$

The chain has five conceptually distinct stages, each resolved by a distinct group of papers.

1. *Discrete-to-continuum.* Q5a establishes the Mosco convergence of the admissibility Dirichlet forms $\mathcal{E}_q$ on $\mathrm{Cay}(G_q, S_q)$ to a one-dimensional shadow operator $L_\Pi = -A\partial_x^2$ on $L^2(\mathbb{R})$. Q5a-O2 closes the two remaining open hypotheses of Q5a ([H-E1] and Conjecture [C]) via spectral atomicity. H2 closes hypothesis [H2].

2. *Dimensional promotion.* Q5b promotes the one-dimensional L_Π to a four-dimensional effective operator L_{eff} on $\mathbb{R}_\tau \times \mathrm{Heis}_3(\mathbb{R})$ via the Bass–Guivarc'h homogeneous dimension $D_{\mathrm{hom}} = 4$ and the Carnot convergence of BFS shells.

3. *Metric extraction.* Q5b Theorem 5.2, made unconditional by Q9's proof of [H-lift], extracts the effective co-metric tensor $g^{\mu\nu} \propto A_{\mu\nu}$ from the principal symbol of L_{eff}. Q5b Theorem 6.1 derives Lorentzian signature $(-,+,+,+)$ from the Born–Infeld admissibility constraint.

4. *Coefficient determination.* Q7 identifies the effective projection space $H_{\mathrm{eff}} = \mathrm{Sym}^2(V_\rho)$ as the unique irreducible three-dimensional $\mathfrak{su}(2)$-module and establishes the bridge rigidity $\phi : \mathrm{Sym}^2(V_\rho) \xrightarrow{\sim} W_{\mathrm{sp}}$. Q8 derives $A_Z = C_{\mathfrak{su}(2)} = 2$ via Casimir rigidity on the central direction. Q10 establishes $A_H \to 2$ under spectral universality [U]. U1 proves [U] with rate $\varepsilon(q) = O(q^{-1/2})$.

5. *Metric closure.* Q11 closes the sole remaining open coefficient $A_\tau = 2$ via temporal Casimir rigidity, giving $g^{\mu\nu} = 2\eta^{\mu\nu}$. W1 closes [H-w].

4 Constituent Papers

4.1 Continuum limit and one-dimensional shadow: Q5a and Q5a-O2

Q5a [6] establishes, conditionally on hypotheses [H1], [H-w], [H-E1], and Conjecture [C], three convergences. (T1) A Hilbert inductive limit $\varinjlim_q \mathcal{C}_q \simeq L^2(\mathbb{R})$, where $\mathcal{C}_q \simeq L^2(\mathbb{Z}/q\mathbb{Z})$ is the Weil representation space at finite q. (T2) Convergence of rescaled Weil generators to the metaplectic generators of $\mathrm{Heis}_3(\mathbb{R})$. (T3) Mosco convergence of the admissibility Dirichlet forms $\mathcal{E}_q$ to the second-order operator $L_\Pi = -A\partial_x^2$ on $L^2(\mathbb{R})$, $A > 0$.

The key conceptual content of Q5a is that the continuum limit is not an additional assumption: it is forced by the coercivity and compactness properties of the admissible forms. Uniform energy bounds prevent spectral mass from escaping to high frequencies; the Mosco limit is the unique operator compatible with spectral stability.

L_Π is a one-dimensional operator. Q5a states explicitly that "the operator L_Π is the one-dimensional operator-theoretic shadow of the homogeneous four-dimensional structure of $\mathrm{Heis}_3(\mathbb{R})$. The identification of L_Π with a four-dimensional effective geometry is the subject of Q5b."

The O-series link is structural: Hypothesis [H-E1] (uniform spectral gap) is equivalent to the stabilisation of δ_{pair} across primes, supported by the O12–O25 campaign. Hypothesis [H-w] (admissibility weight convergence $a_q(s) \to A > 0$) is proved in W1 [17].

Q5a-O2 [13] resolves the two remaining open hypotheses of Q5a. It proves that the admissible sector $\mathrm{ran}(\Pi_q)$ is spectrally atomic: it is spanned by three pure Fourier modes per conjugate pair, with macroscopic frequencies $\xi_c^{(q)}/q \to \omega_c \neq 0$. From this structure, coercivity at the optimal diffusive scale follows without any Nash or Rellich inequality, and Mosco compactness follows from Bolzano–Weierstrass in $\mathbb{C}^2$. Hypotheses [H-E1] and [C] become unconditional theorems on the admissible sector.

4.2 Carnot geometry and Lorentzian signature: Q5b

Q5b [2] delivers three results that together promote the one-dimensional shadow L_Π to a four-dimensional effective geometry.

Carnot convergence (Theorem 3.2; structural). The BFS shell stratification $\{S_n\}$ of $G_q = \mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ converges, in the pre-saturation regime $n \ll q$, to the Carnot–Carathéodory sphere foliation of $\mathrm{Heis}_3(\mathbb{R})$. The homogeneous dimension $D_{\mathrm{hom}} = 4$ (Bass–Guivarc'h) gives the limiting geometry the spectral and volume-growth properties of a four-dimensional space. This result is structural: it follows from Pansu's asymptotic cone theorem and requires no new hypothesis.

Metric extraction (Theorem 5.2; unconditional via Q9 [11]). Under [H-lift] — proved in Q9 as a theorem rather than a hypothesis — L_Π is the image, under the Schrödinger representation π_1, of the kinetic sector of the sub-Riemannian Laplacian $\Delta_H = \tilde{X}^2 + \tilde{Y}^2$ on $\mathrm{Heis}_3(\mathbb{R})$, restricted to the admissible sector. The effective co-metric $g^{\mu\nu} \propto A_{\mu\nu}$ is extracted from the principal symbol of the resulting effective operator L_{eff} on $\mathbb{R}_\tau \times \mathrm{Heis}_3(\mathbb{R})$.

Lorentzian signature (Theorem 6.1; conditional on Theorem 5.2). The Born–Infeld admissibility constraint forces signature $(-,+,+,+)$ by exclusion: Riemannian signature is excluded by the finite-capacity bound of A4; ultra-hyperbolic signature fails to admit a globally well-posed initial value problem; degenerate signature is excluded by sub-principal completion; parabolic signature implies a characteristic direction incompatible with monotonic projective ordering.

After normalising the maximal admissible propagation speed to $c = 1$, the effective metric is fixed to $g^{\mu\nu} = \mathrm{diag}(-1, +1, +1, +1)$ in the homogeneous isotropic regime.

4.3 Effective spacetime output: Q6b

Q6b [3] is the integrative output of the sub-programme. Its role is not to re-derive the geometric convergence theorems of Q5a–Q5b, but to assemble the chain into the macroscopic effective-spacetime statement:

$$\Pi_q \longrightarrow L_{\mathrm{eff}} \longrightarrow g^{\mu\nu} \longrightarrow G_{\mu\nu}.$$

Three results are obtained. (i) The Schwarzschild metric as the unique stationary, spherically symmetric exterior solution compatible with the admissibility flow, obtained from flux conservation within the effective geometry. (ii) A structural interpretation of the Schwarzschild horizon as a degeneracy of the principal symbol of L_{eff}, not a singularity of the underlying admissible structure. (iii) The identification of the Einstein equations as consistency conditions of the emergent geometry, via the spectral entropy functional of the Gravity paper.

Q6b closes the chain $\Pi_q \to L_{\mathrm{eff}} \to g^{\mu\nu} \to G_{\mu\nu}$ under the Q5a–Q5b hypotheses. With [H-lift] proved in Q9, the results of Q6b are unconditional with respect to [H-lift]. The remaining conditionality stems from the Q5a Mosco hypotheses [H1], [H-w], [H-E1], and [C]; of these, [H-w] is proved in W1, and [H-E1] and [C] are closed in Q5a-O2.

Remark 3 (Status of Q6b)**.** Q6b is listed as published in the programme. Its status as *integrative output* means that it assembles rather than proves: it inherits the metric from Q5b and passes the Einstein equations to the Gravity paper. It should not be cited as the primary source for either the metric or the Einstein equations; those primary sources are Q5b and the Gravity paper respectively.

4.4 Spatial, vertical, horizontal, and temporal coefficients: Q7–Q11

Papers Q7–Q11 determine the four numerical entries of the effective metric by distinct representation-theoretic arguments, all converging to the same value $A = 2$.

Q7 [15] establishes three structural results. First, $\mathfrak{su}(2) \not\cong \mathfrak{heis}_3(\mathbb{R})$ as Lie algebras, so the bridge $\phi : H_{\mathrm{eff}} \to W_{\mathrm{sp}}$ operates at the representation level. Second, the effective projection space $H_{\mathrm{eff}} = \mathrm{Sym}^2(V_\rho)$ carries the unique irreducible three-dimensional $\mathfrak{su}(2)$-module (from O29). The three spatial dimensions are the representation-theoretic image of $\mathrm{Im}\,\mathbb{H} \cong \mathfrak{su}(2)$, confirming the causal direction: $\Sigma_c(n_3) = 3 \Rightarrow \dim \mathrm{Im}\,\mathbb{H} = 3 \Rightarrow$ three spatial directions. Third, the bridge ϕ is unique up to positive scalar (Rigidity Lemma, Q7 Lemma 4.3), so the metric coefficients are uniquely determined. The no-cross-terms condition $k_X k_Z = k_Y k_Z = 0$ is proved analytically (Proposition 6.1).

Q9 [11] proves hypothesis [H-lift] as a theorem by the method of generator suppression: the modulation generator is suppressed at rate $O(q^{-1/2})$ on the admissible sector, and coercivity of the limit form forces $A_H > 0$ independently of bridge existence. This makes Q5b Theorems 5.2 and 6.1 unconditional with respect to [H-lift].

Q8 [8] establishes $A_Z = C_{\mathfrak{su}(2)} = 2$ for the central (vertical) direction via Casimir rigidity. The Heisenberg commutator $[\tilde{X}, \tilde{Y}] = \tilde{Z}$ has a unique algebraic source; under the equivariant bridge ϕ, its coefficient is constrained to be proportional to the unique $\mathfrak{su}(2)$-invariant quadratic form on $\mathrm{Sym}^2(V_\rho)$, namely the Casimir $C_{\mathfrak{su}(2)} = 2 \cdot \mathrm{Id}$. The normalisation is fixed with no free scalar. This closes Q5b open problem O2.

Q10 [9] proves $A_H \to 2$ as $q \to \infty$ under the O-series spectral universality hypothesis [U]. The argument runs: character-independence of the admissibility weights forces $\mathfrak{su}(2)$-isotropy of the effective quadratic form on $H_{\mathrm{eff}} = \mathrm{Sym}^2(V_\rho)$; the unique $\mathfrak{su}(2)$-invariant quadratic form on this spin-1 module has eigenvalue 2 (Casimir). Numerical confirmation from the O25 campaign gives $|A_H(q) - 2| \approx 0.264q^{-0.44}$ for $q \in \{61, 101, 151\}$.

U1 [16] proves hypothesis [U] with rate $\varepsilon(q) = O(q^{-1/2})$ from the Weil-generator Lipschitz bound and the BFS–Carnot–Carathéodory convergence. This closes [U] as a theorem rather than a structural assumption.

Q11 [14] closes the sole remaining open coefficient A_τ. The cascade increment operator T internalises no new irreducible representation; by Schur's lemma, the τ-sector carries the trivial SU(2)-representation and the temporal quadratic form must be SU(2)-invariant. Since T is the same cascade that fixes the spatial bridge normalisation (Q8), no independent normalisation arises: $A_\tau = C_{\mathfrak{su}(2)} = 2$. This gives the isotropic Lorentzian co-metric:

$$g^{\mu\nu} = 2\eta^{\mu\nu}, \qquad \eta = \mathrm{diag}(-1, +1, +1, +1). \tag{2}$$

The conformal factor of 2 is not a choice of units; it is the eigenvalue of the $\mathfrak{su}(2)$-Casimir on the spin-1 module $\mathrm{Sym}^2(V_\rho)$, governing all four metric components uniformly. After absorbing $\Omega^2 = 2$ into the normalisation of the propagation speed, the effective metric is exactly Minkowski.

4.5 Closure papers: W1 and H2

W1 [17] proves hypothesis [H-w] (convergence of the admissibility weights $a_q(s) \to A > 0$) as a structural consequence of the fibre-invariance of admissible observables, made quantitative by U1. After W1, the open Q5a hypotheses reduce to [H1], [H-E1], and [C].

H2 [12] proves hypothesis [H2] (strong convergence of the rescaled Weil generators $\hat{X}_q \to x$ and $\hat{P}_q \to -i\partial_x$) via a quantitative Poisson aliasing lemma and a discrete Sobolev identity. Together with Q5a-O2, H2 completes the closure of all Q5a hypotheses on the admissible sector.

Table 1: Summary of constituent papers by internal phase. Status: P = proved; S = structural; C = conditional on Q5a hypotheses.

Phase	Papers	Central output	Status
Discrete-to-continuum	Q5a, Q5a-O2, H2	$L_\Pi = -A\partial_x^2$; [H-E1],[C],[H2] closed	P/S
Dimensional promotion	Q5b	$D_{\mathrm{hom}} = 4$; Carnot convergence	S
Metric extraction	Q5b + Q9	$g^{\mu\nu} \propto A_{\mu\nu}$; signature $(-,+,+,+)$	P
Coefficient determination	Q7, Q8, Q10, U1	$A_Z = A_H = 2$	P/S
Metric closure	Q11, W1	$A_\tau = 2$; $g^{\mu\nu} = 2\eta^{\mu\nu}$	S/P
Integrative output	Q6b	$\Pi_q \to L_{\mathrm{eff}} \to g^{\mu\nu} \to G_{\mu\nu}$	C

5 Inputs from Upstream Results

The emergent geometry sub-programme takes the following inputs from Branch I and from Presentation Note 1 (spectral admissibility).

1. **Weil representation on $L^2(\mathbb{Z}/q\mathbb{Z})$.** The identification of the admissible fibre as V_ρ is a theorem of Foundation M and HeisenbergStructure [7, 1].

2. **Born–Infeld admissibility constraint.** The bounded-flux condition $A_n \leq c_\chi/\sqrt{\lambda_n}$ is the Q5a starting point. Its origin in the BI action is established in Branch I.

3. **Spin-$\frac{1}{2}$ sector and $\Sigma_c(n_3) = 3$.** From Presentation Note 1 (O23, O27, O29): the admissible sector is $V_\rho \cong \mathbb{C}^2$; the threshold dimension is $\Sigma_c(n_3) = 3$ by quaternionic minimality; $\operatorname{Im}\mathbb{H} \cong \mathfrak{su}(2)$ is the three-dimensional spatial algebra. These are the inputs to Q7's identification of $H_{\text{eff}} = \operatorname{Sym}^2(V_\rho)$ as the effective projection space.

4. **O-series universality data.** The stabilisation of $\delta_{\text{pair}}(q)$ across the O12–O25 campaign provides the numerical support for [H-E1] and for the spectral universality [U] required by Q10.

6 Outputs to Downstream Branches

Table 2 lists the principal outputs of the sub-programme and their downstream consumers.

Table 2: Principal outputs and downstream consumers.

Output	Consumer	Status
$g^{\mu\nu} = 2\eta^{\mu\nu}$	Gravity, Q12, Q13	S (Q11)
Lorentzian signature $(-, +, +, +)$	All Branch III	P (Q5b + Q9)
L_{eff} on $\mathbb{R}_\tau \times \text{Heis}_3(\mathbb{R})$	Q6b, Gravity, Q12	C
$H_{\text{eff}} = \operatorname{Sym}^2(V_\rho) \cong \mathbb{C}^3$	Q7, Q8, Q14	P
Three spatial directions from $\operatorname{Im}\mathbb{H}$	Q7, gauge notes	P
Schwarzschild exterior metric	Q6b	C
Einstein equations as consistency conditions	Q6b, Gravity	C
ℓ_{sp} (spectral length scale)	Gravity [5]	S

7 Status of Results

7.1 Proved results

1. $D_{\text{hom}} = 4$ (Q5b Theorem 2.1): the Bass–Guivarc'h theorem for $\text{Heis}_3(\mathbb{R})$; the homogeneous dimension is 4.

2. *Carnot convergence* (Q5b Theorem 3.2, following Pansu): the BFS shell stratification of G_q converges to the Carnot–Carathéodory sphere foliation of $\text{Heis}_3(\mathbb{R})$ in the pre-saturation regime.

3. [*H-lift*] (Q9 Theorem 1.2): the admissibility filter concentrates on the kinetic sector of $d\pi_1(\Delta_H)$ at rate $O(q^{-1/2})$, making L_Π the kinetic shadow of Δ_H without independent hypothesis.

4. *Metric extraction* (Q5b Theorem 5.2, unconditional via Q9): the effective co-metric $g^{\mu\nu} \propto A_{\mu\nu}$ is extracted from the principal symbol of L_{eff} under the Q5a hypotheses.

5. *Lorentzian signature* (Q5b Theorem 6.1): signature $(-,+,+,+)$ is forced by the Born–Infeld admissibility constraint via exclusion of all other signature classes.

6. *Bridge rigidity* (Q7 Lemma 4.3): the bridge $\phi : \text{Sym}^2(V_\rho) \xrightarrow{\sim} W_{\text{sp}}$ is unique up to positive scalar.

7. *No-cross-terms* (Q7 Proposition 6.1): the cross terms $k_X k_Z = k_Y k_Z = 0$ are proved analytically from $[\mathcal{F}_c, L_{\text{Weil}}] = 0$.

8. $A_Z = 2$ (Q8 Theorem 1.1): Casimir rigidity of the central direction; $A_Z = C_{\mathfrak{su}(2)}|_{\text{Sym}^2(V_\rho)} = 2 \cdot \text{Id}$ with no free normalisation scalar.

9. *Spectral universality [U]* (U1): $\varepsilon(q) = O(q^{-1/2})$; proved from the Weil-generator Lipschitz bound.

10. $A_H \to 2$ (Q10 Theorem 1.1): character-independence forces $\mathfrak{su}(2)$-isotropy of the effective form; the Casimir gives value 2 in the limit.

11. [*H-w*] (W1): admissibility weight convergence $a_q(s) \to A > 0$ proved as a structural consequence of fibre-invariance, quantified by U1.

12. [*H2*] (H2): strong convergence of rescaled Weil generators via Poisson aliasing lemma and discrete Sobolev identity.

13. [*H-E1*] *and* [*C*] *on the admissible sector* (Q5a-O2): proved via spectral atomicity of $\text{ran}(\Pi_q)$.

14. $A_\tau = 2$ (Q11 Theorem 1.1): temporal Casimir rigidity; cascade internalization lemma, Schur rigidity, and Q8 normalisation close A_τ.

15. *Metric closure* $g^{\mu\nu} = 2\eta^{\mu\nu}$ (Q11 Corollary 6.1): combination of $A_\tau = 2$ (Q11), $A_H = 2$ (Q10), $A_Z = 2$ (Q8), and Q5b Theorem 6.1.

7.2 Structural results

1. $H_{\text{eff}} = \text{Sym}^2(V_\rho)$ (Q7 Proposition 4.1, from O29): the effective projection space is the unique irreducible three-dimensional $\mathfrak{su}(2)$-module.

2. *Three spatial dimensions from* $\text{Im}\,\mathbb{H}$ (Q7, O23): the three spatial directions are the image of $\text{Im}\,\mathbb{H} \cong \mathfrak{su}(2)$; three-dimensionality is not postulated.

3. *Schwarzschild exterior metric* (Q6b): flux conservation within the effective geometry forces the Schwarzschild metric as the unique stationary spherically symmetric exterior solution; conditional on Q5a hypotheses.

4. *Einstein equations as consistency conditions* (Q6b): the Einstein–Hilbert action is identified as the spectral entropy functional of the Gravity paper; conditional on Q5a hypotheses.

5. A_Z *is not a free parameter* (Q8 §6.3): the value $A_Z = 2$ is fixed by the Casimir of the unique irreducible $\mathfrak{su}(2)$-module of dimension 3; it is a structural identity, not a fitted constant.

7.3 Numerical evidence

1. $|A_H(q)-2| \approx 0.264q^{-0.44}$ (Q7, Q10): confirmed from the O25 campaign for $q \in \{61, 101, 151\}$; consistent with convergence to 2 with rate $O(q^{-1/2})$.

2. *Mosco tightness (Conjecture [C])* (Q5a): spectral tail mass satisfies $\mathcal{E}_q(q/3)/\|f_q\|^2 < 2.1 \times 10^{-5}$ for $q \in \{29, 61, 101, 151\}$, with values decreasing toward machine precision. Closed analytically on the admissible sector by Q5a-O2.

7.4 Open hypotheses

1. *Hypothesis [H1] on the full function space.* Q5a Hypothesis [H1] (embedding $\iota_q : \mathcal{C}_q \hookrightarrow L^2(\mathbb{R})$ with uniform control) is not needed in full generality on the admissible sector (Q5a-O2), but its extension to the full L^2 context remains open. This does not block any downstream result.

2. *Bridge existence at finite q* (Q8-O1). Q7 Conjecture 4.7 / Q8-O1: does an explicit $\mathfrak{su}(2)$-equivariant isomorphism $\phi_q : \mathrm{Sym}^2(V_\rho) \xrightarrow{\sim} W_{\mathrm{sp}}$ exist at each prime q, not only in the asymptotic limit? The programme consequences ($A_Z = 2$, $A_H \to 2$, $A_\tau = 2$) are established without it; this is a secondary structural open problem.

8 Open Deliverables

Two open items define the boundary of the sub-programme as of this note.

Open deliverable 1: Bridge existence at finite q. The asymptotic results ($A_Z = A_H = A_\tau = 2$) and the metric closure $g^{\mu\nu} = 2\eta^{\mu\nu}$ are established in the $q \to \infty$ limit. The finite-q equivariant bridge ϕ_q — required for a fully explicit representation-level identification at each prime — remains an open structural problem. Its resolution would strengthen the narrative but would not change any published result.

Open deliverable 2: [H1] on the full L^2 space. Q5a-O2 closes [H1] on the admissible sector. The full-space version is not needed by any result in the programme and is listed here for completeness.

9 Conclusion

The emergent geometry sub-programme derives the effective Lorentzian metric $g^{\mu\nu} = 2\eta^{\mu\nu}$ from the admissibility filter Π_q acting on the Weil representation of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$.

The chain has five stages, each resolved by distinct papers: the discrete-to-continuum Mosco limit (Q5a, Q5a-O2, W1, H2); the dimensional promotion via Carnot geometry (Q5b); the metric extraction and signature selection (Q5b + Q9); the determination of the metric coefficients A_Z, A_H, A_τ via Casimir rigidity (Q7–Q11, U1); and the assembly into the effective spacetime statement (Q6b). All four metric coefficients equal 2, a single representation-theoretic datum: the eigenvalue of the $\mathfrak{su}(2)$-Casimir on the spin-1 module $\mathrm{Sym}^2(V_\rho)$.

The sub-programme closes the reconstruction problem: $g^{\mu\nu}$ is not postulated but derived. The Lorentzian signature is not postulated but excluded by elimination. The value of the conformal factor is not fitted but forced by the Casimir.

This note does not address the dynamics of $g^{\mu\nu}$. The question of why the effective metric satisfies the Einstein equations is handled by the spectral gravity sub-programme (Presentation Note 4), which takes $g^{\mu\nu} = 2\eta^{\mu\nu}$ as its starting point.

References

[1] J. Beau. Admissible Non-Injective Transitions as the Primitive of Physical Description, 2026. Cosmochrony Foundation Paper "M", doi:10.5281/zenodo.20258438.

[2] J. Beau. BFS Shell Stratification and the Emergence of Four-Dimensional Lorentzian Geometry, 2026. spectral admissibility subprogram "Q5b" paper, doi:10.5281/zenodo.20277381.

[3] J. Beau. Effective Spacetime Geometry from Admissible Non-Injective Projection, 2026. spectral admissibility subprogram "Q5b" paper, doi:10.5281/zenodo.20257944.

[4] J. Beau. Gauge Structure from Admissible Non-Injective Projection, 2026. spectral admissibility subprogram "Q5a" paper, doi:10.5281/zenodo.19655294.

[5] J. Beau. Infrared Einstein Response from a Renormalized Spectral Entropy Functional, 2026. Cosmochrony companion paper H, doi:10.5281/zenodo.18818721.

[6] J. Beau. Large-q Limit of the Admissible Fibre: Representational and Operator Convergence, 2026. spectral admissibility subprogram "Q5a" paper, doi:10.5281/zenodo.19642369.

[7] J. Beau. Non-Factorisability and the Emergence of Heisenberg Structure from Admissibility Constraints, 2026. spectral admissibility subprogram paper, doi:10.5281/zenodo.19635395.

[8] J. Beau. Sub-Principal Symbol of the Effective Operator and Casimir Rigidity of the Central Direction, 2026. "Q8" paper, doi:10.5281/zenodo.19879909.

[9] Jérôme Beau. Asymptotic $\mathfrak{su}(2)$-Isotropy of the Effective Quadratic Form and the Identification $A_H = 2$, 2026. "Q10" paper, doi:10.5281/zenodo.19880900.

[10] Jérôme Beau. Fermionic Matter and Chirality from Projective Dirac Admissibility, 2026. "Q14" paper, doi:10.5281/zenodo.20262081.

[11] Jérôme Beau. Proof of the Lifting Hypothesis [H-lift]: Kinetic-Sector Identification via Generator Suppression, 2026. "Q9" paper, doi:10.5281/zenodo.19880574.

[12] Jérôme Beau. Semiclassical Consistency of the Weil Representation on $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$: Quantitative Sinc Embedding, Generator Convergence, and Aliasing Control, 2026. "H2" paper, doi:10.5281/zenodo.19962051.

[13] Jérôme Beau. Spectral Atomicity of the Admissible Sector: Scaled Coercivity and Mosco Compactness without Nash Inequalities, 2026. "Q5a-O2" paper, doi:10.5281/zenodo.19915951.

[14] Jérôme Beau. Temporal Casimir Rigidity and Closure of the Effective Metric: Identification of $A_\tau = 2$, 2026. "Q11" paper, doi:10.5281/zenodo.20098387.

[15] Jérôme Beau. Three Admissible Directions and Three Spatial Dimensions: A Structural Bridge Candidate, 2026. "Q7" paper, doi:10.5281/zenodo.19802123.

[16] Jérôme Beau. Uniform Spectral Universality for Weil Fingerprint Energies: Proof of [U] with Rate $O(q^{-1/2})$, 2026. "U1" paper, doi:10.5281/zenodo.19881146.

[17] Jérôme Beau. Weight Stabilisation of the Admissible Dirichlet Form: Proof of Hypothesis [H-w] from Spectral Universality, 2026. "W1" paper, doi:10.5281/zenodo.19886319.